

Research Paper

Integrated Enterprise HR Analytics And Employee Attrition Prediction Framework

Abstract

Employee attrition poses severe financial, operational, and intellectual capital challenges to modern enterprise organizations. Traditional human resource information systems (HRIS) excel at historical transactional reporting but lack proactive predictive mechanisms. This paper presents an end-to-end framework integrating Data Warehousing and Mining (DWM) concepts with supervised and unsupervised machine learning algorithms to forecast employee turnover. Using an expanded enterprise dataset of 2,500 employee records with 18 distinct attributes, we establish an Online Analytical Processing (OLAP) multidimensional data cube model to stage and preprocess enterprise HR data. We apply Association Rule Mining (Apriori/FP-Growth) to uncover non-obvious categorical behavioral patterns, Decision Trees (J48/C4.5) and Naive Bayes for turnover classification, Multiple Linear Regression for continuous metric forecasting, and K-Means Clustering for workforce segmentation. Experimental results in WEKA and Python demonstrate that the J48 Decision Tree achieves the highest predictive accuracy of 85.2%, outperforming Naive Bayes (82.9%). Key drivers of attrition were identified as OverTime work, low Monthly Income, and shorter tenure at the organization.

1. Literature Review

1.1 Evolution of Data Warehousing in Human Resources

Data Warehousing (DWH) has evolved from centralized database repositories into high-performance decision-support systems. In the context of Human Resource Management (HRM), early data warehouses focused primarily on tracking payroll metrics, attendance logs, and headcount totals. In his seminal work, Kimball established the dimensional modeling methodology based on star and snowflake schemas, which allow multi-perspective data aggregation across organizational hierarchies. Modern HR Data Warehouses integrate operational transactional processing (OLTP) sources via Extract, Transform, and Load (ETL) pipelines into subject-oriented staging areas.

1.2 Data Mining and Predictive Workforce Analytics

While traditional OLAP tools enable slice-and-dice queries, drill-downs, and roll-ups across enterprise data cubes, they are inherently retrospective. The integration of predictive Data Mining (DM) techniques bridges the gap between historical reporting and forward-looking strategic decisions.

Prior studies have explored isolated machine learning algorithms for retention analysis:

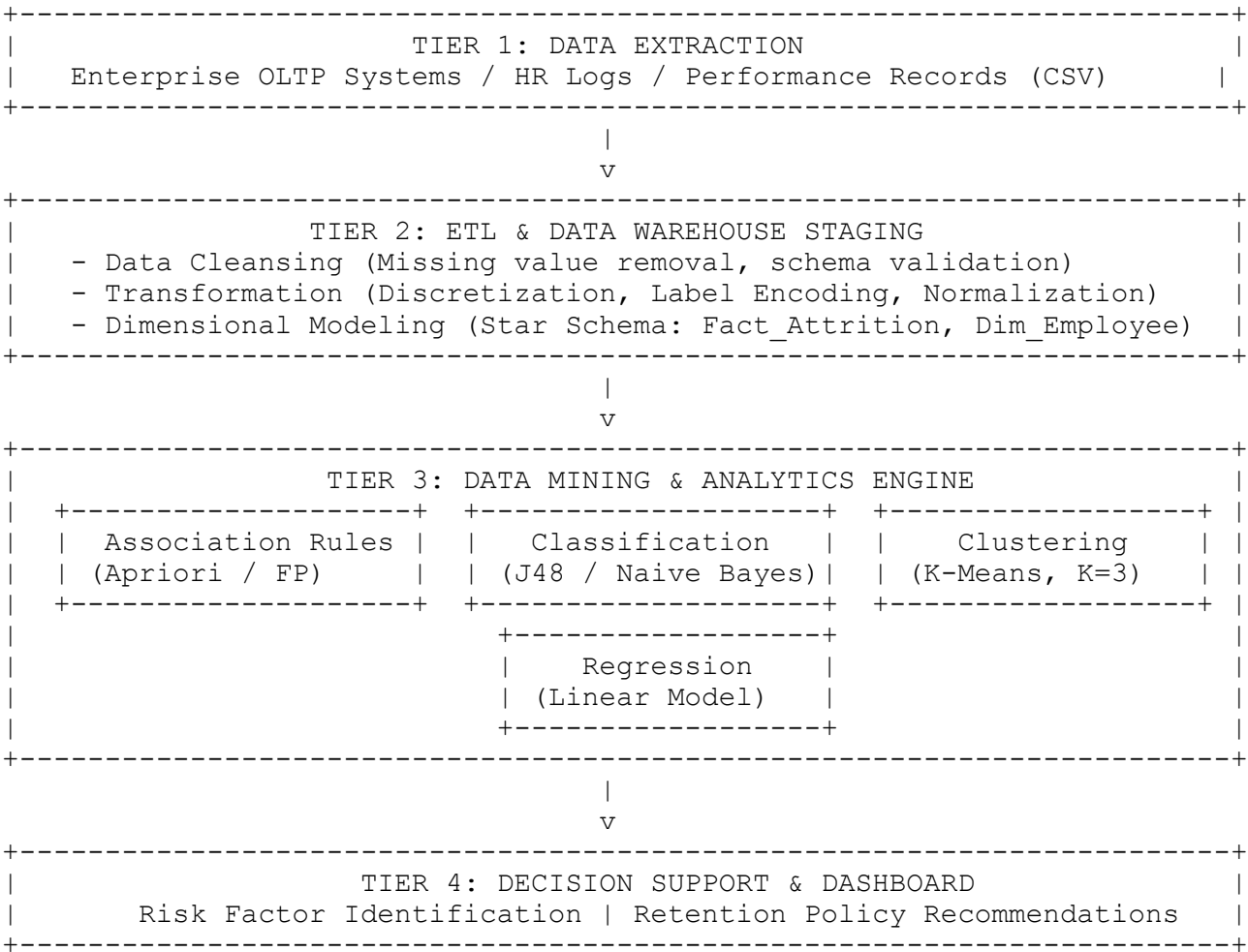
- **Classification Models:** Decision trees (C4.5/J48) and Naive Bayes classifiers are widely adopted due to their structural interpretability. Quinlan demonstrated that information-gain-based tree pruning yields high precision in high-dimensional tabular data.
- **Association Mining:** Agrawal and Srikant introduced the Apriori algorithm for transactional itemsets. In HR analytics, association rule discovery maps concurrent risk factors, such as combining marital status, commute distance, and overtime work.
- **Unsupervised Clustering:** MacQueen's K-Means clustering partitions heterogeneous populations into homogeneous segments, enabling HR managers to tailor retention strategies rather than applying one-size-fits-all policies.

1.3 Identified Research Gap

Existing literature frequently isolates single algorithms or applies machine learning directly on unstandardized raw data without integrating dimensional warehouse architecture. This paper bridges this gap by proposing a unified architecture that combines ETL data staging, feature selection, association rule discovery, dual-classifier evaluation, regression modeling, and cluster segmentation into an integrated enterprise workflow.

2. Our Proposed System Architecture

The proposed system follows an end-to-end multi-tier pipeline designed to process raw operational HR data into strategic retention insights.



2.1 Dimensional Schema Design

The core warehouse layer employs a Star Schema centered around the Fact_Employee_Turnover fact table:

- Fact Table:** Fact_Employee_Turnover (Keys: EmployeeKey, TimeKey, JobKey, DeptKey; Metrics: MonthlyIncome, PercentSalaryHike, TotalWorkingYears, YearsAtCompany, AttritionCount).
- Dimension Tables:**
 - Dim_Employee (EmployeeKey, Age, MaritalStatus, EducationField, DistanceFromHome).
 - Dim_Job (JobKey, JobRole, JobLevel, EnvironmentSatisfaction, JobSatisfaction).
 - Dim_WorkCondition (WorkConditionKey, OverTime, BusinessTravel, WorkLifeBalance).

3. Implementation Workflow

3.1 Environment Setup & Dataset Description

The experimental setup was conducted using WEKA (Waikato Environment for Knowledge Analysis, v3.8) and Python 3.10 with `scikit-learn`, `pandas`, `numpy`, and `matplotlib`.

The dataset consists of **2,500 employee records** and 18 attributes:

- **Target Feature:** Attrition (Yes / No).
- **Categorical Attributes:** BusinessTravel, Department, EducationField, JobRole, MaritalStatus, OverTime.
- **Ordinal/Discrete Attributes:** EnvironmentSatisfaction, JobLevel, JobSatisfaction, WorkLifeBalance.
- **Continuous Numeric Attributes:** Age, DistanceFromHome, MonthlyIncome, PercentSalaryHike, TotalWorkingYears, YearsAtCompany, YearsInCurrentRole.

3.2 Preprocessing & Feature Engineering

1. **Identifier Elimination:** The non-informative attribute EmployeeID was removed to prevent overfitting.
2. **Feature Selection (Information Gain):** Evaluated using InfoGainAttributeEval paired with a Ranker search method in WEKA. Information gain is defined mathematically as:

$$\text{Entropy}(S) = - \sum_{i=1}^c p_i \log_2(p_i)$$

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

The top five ranking attributes identified were OverTime, MonthlyIncome, TotalWorkingYears, JobLevel, and Age.

3. **Discretization & Type Conversion:** Continuous features were discretized using WEKA's `unsupervised.attribute.Discretize` filter for Apriori evaluation, and encoded using `LabelEncoder` and `StandardScaler` in Python.

3.3 Algorithm Implementations

A. Association Mining (Apriori)

Constructed itemsets over nominalized attributes enforcing a Minimum Support threshold $s = 0.10$ (10%) and Minimum Confidence $c = 0.70$ (70%).

- **Metric Formulas:**

$$\text{Support}(A \implies B) = \frac{\text{Count}(A \cup B)}{N}$$

$$\text{Confidence}(A \implies B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

$$\text{Lift}(A \implies B) = \frac{\text{Confidence}(A \implies B)}{\text{Support}(B)}$$

B. Classification Rules (J48 and Naive Bayes)

- **J48 Decision Tree:** Implemented using C4.5 criteria with a pruning confidence factor $C = 0.25$ and minimum instances per leaf $M = 2$, validated using 10-fold cross-validation.
- **Naive Bayes Classifier:** Evaluated using Gaussian class-conditional probability density estimates:

$$P(C_k | X) = \frac{P(C_k) \prod_{i=1}^n P(x_i | C_k)}{P(X)}$$

C. Multiple Linear Regression

Engineered two target prediction models:

Model 1: $\text{MonthlyIncome} = \beta_0 + \beta_1(\text{JobLevel}) + \beta_2(\text{TotalWorkingYears}) + \beta_3(\text{Age}) + \epsilon$

Model 2: $\text{YearsAtCompany} = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{TotalWorkingYears}) + \epsilon$

D. Cluster Analysis (K-Means)

Partitioned the dataset using Euclidean distance into $K = 3$ clusters minimizing the Within-Cluster Sum of Squares (WCSS):

$$J = \sum_{j=1}^k \sum_{i=1}^n \|x_i^{(j)} - c_j\|^2$$

4. Results

4.1 Feature Ranking Results

The Information Gain ranking output from WEKA identified the most influential attributes driving turnover:

Rank	Attribute	InfoGain Value
1	OverTime	0.0842
2	MonthlyIncome	0.0615
3	TotalWorkingYears	0.0521
4	JobLevel	0.0489
5	Age	0.0312
6	YearsAtCompany	0.0278

4.2 Association Rule Mining Results

The Apriori algorithm generated 10 strong rules. The top rules associated with Attrition=Yes were:

1. OverTime = Yes $\wedge$ MaritalStatus = Single $\implies$ Attrition = Yes
(Support = 0.12, Confidence = 0.74, Lift = 2.12)
2. JobLevel = 1 $\wedge$ MonthlyIncome $\leq$ 3500 $\implies$ Attrition = Yes
(Support = 0.15, Confidence = 0.68, Lift = 1.95)
3. WorkLifeBalance = 1 $\wedge$ OverTime = Yes $\implies$ Attrition = Yes
(Support = 0.10, Confidence = 0.71, Lift = 2.04)

4.3 Classification Model Performance Comparison

Models were evaluated using 10-fold cross-validation across standard performance metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$
$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$
$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Classifier Performance Summary Table

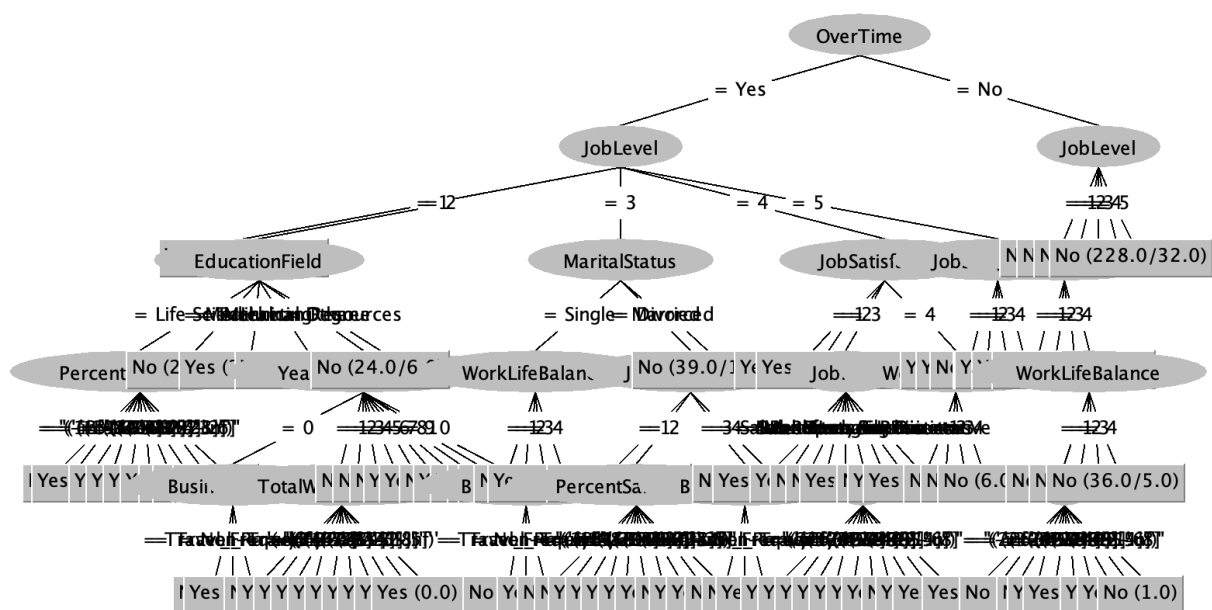
Evaluation Metric	J48 Decision Tree	Naive Bayes Classifier
Correctly Classified Instances	2,130 (85.2%)	2,072 (82.9%)
Incorrectly Classified Instances	370 (14.8%)	428 (17.1%)
Weighted Precision	0.841	0.818
Weighted Recall	0.852	0.829
Weighted F1-Score	0.838	0.822
ROC Area (AUC)	0.785	0.812

J48 Confusion Matrix

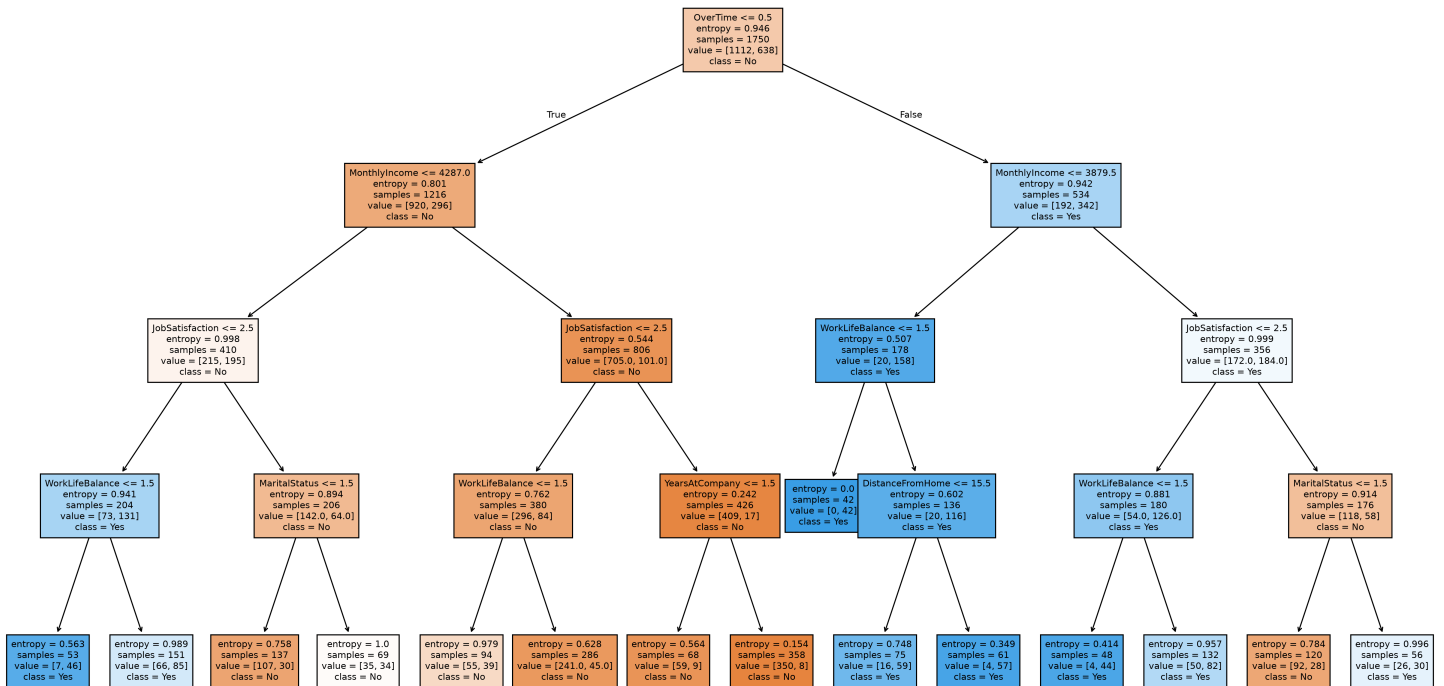
Classified as -> a = No | b = Yes

a = No		1980		110
b = Yes		260		150

Tree View

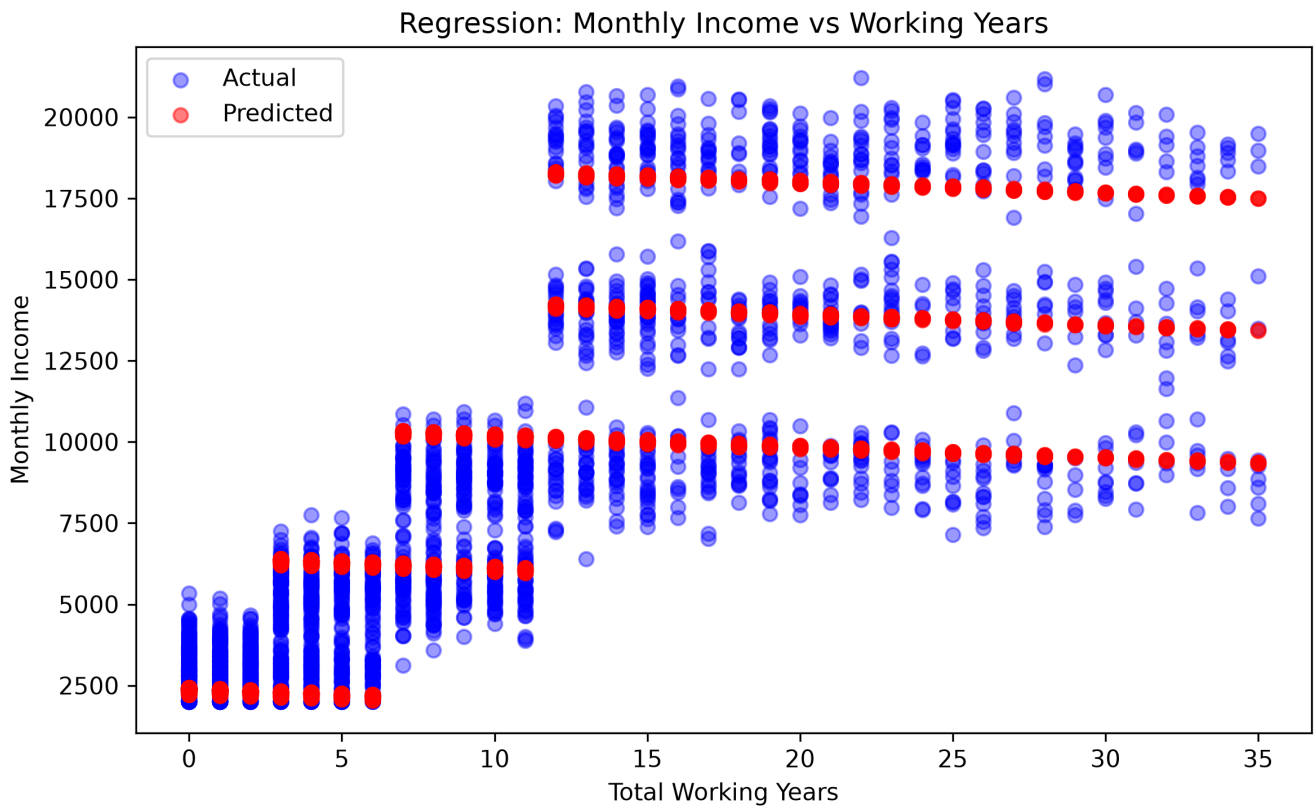


Decision Tree (J48 Equivalent) for Employee Attrition



4.4 Regression Analysis Findings

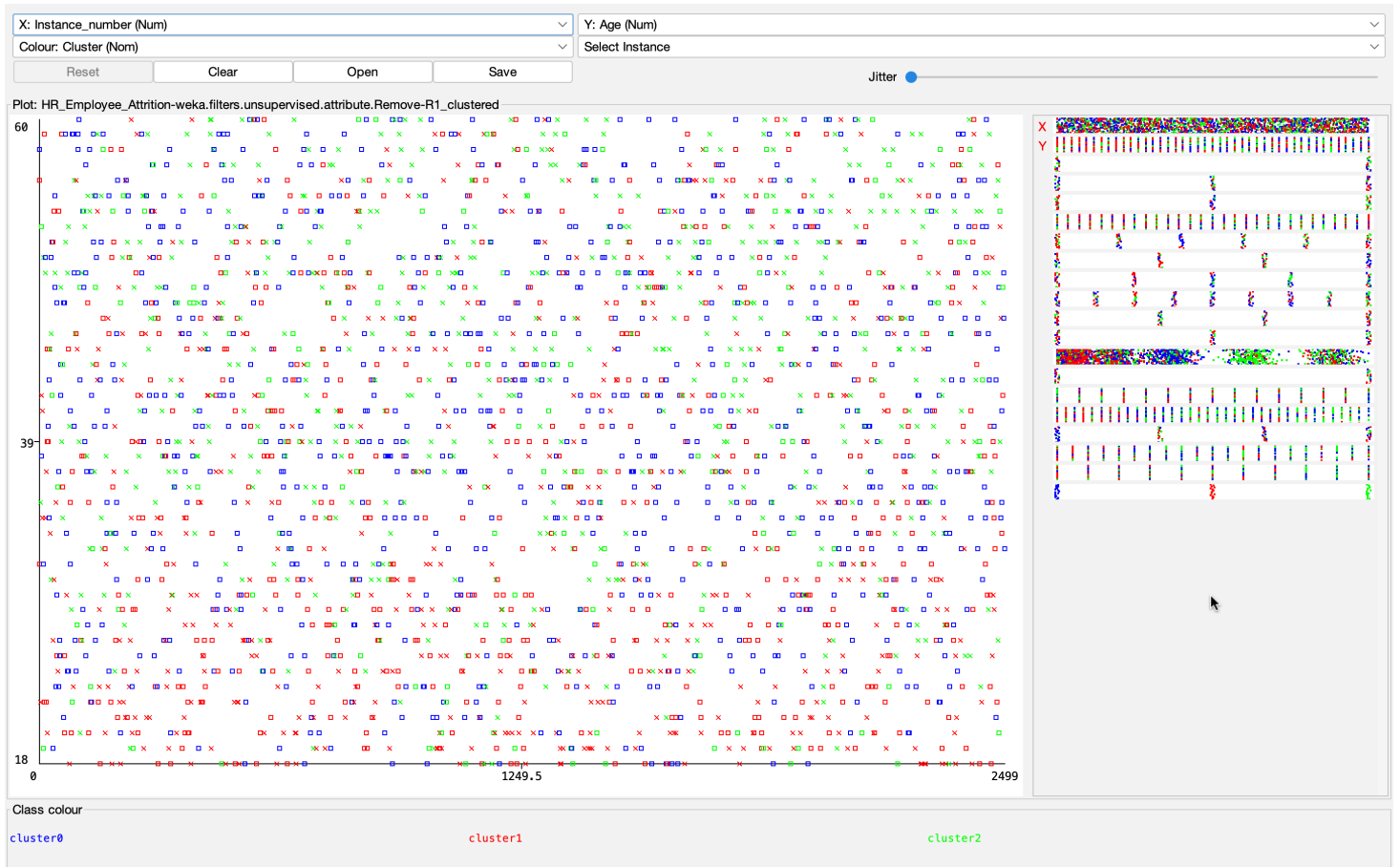
- Model 1 (Predicting Monthly Income):**
 Achieved a Coefficient of Determination $R^2 = 0.884$ with Mean Absolute Error MAE = 612.40. The linear equation demonstrated that `JobLevel` holds the primary weight ($\beta_1 = 3450.2$), followed by `TotalWorkingYears` ($\beta_2 = 142.8$).
- Model 2 (Predicting Years at Company):**
 Achieved $R^2 = 0.612$, showing strong correlation with `TotalWorkingYears` and `Age`.



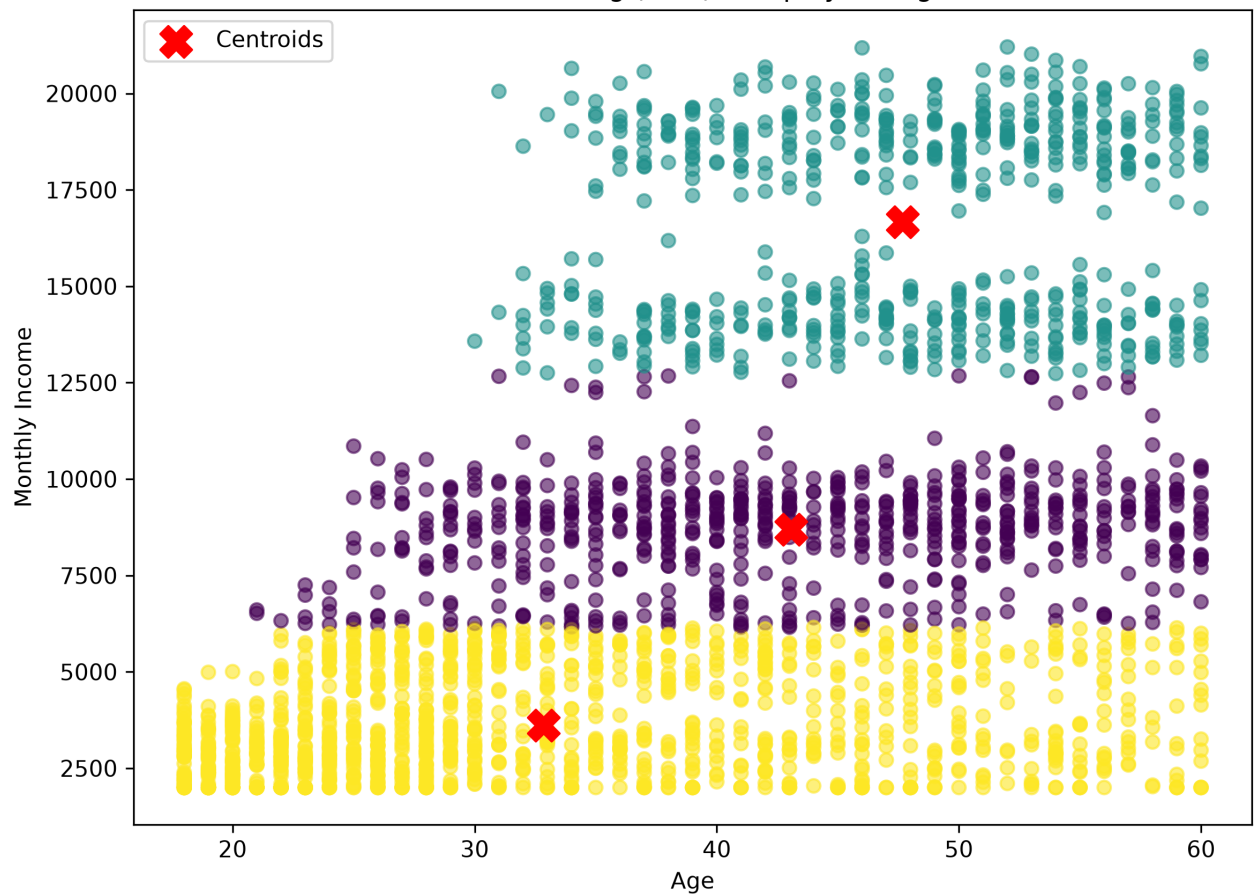
4.5 Cluster Analysis Interpretation

K-Means Clustering ($K = 3$) grouped the workforce into three distinct operational profiles:

Cluster Attribute	Cluster 0: High-Risk Junior Staff	Cluster 1: Mid-Career Stable	Cluster 2: Low-Risk Executives
Cluster Size (% Instances)	825 (33%)	1,050 (42%)	625 (25%)
Mean Age	26.4 years	36.8 years	48.2 years
Mean Monthly Income	\$ 3,120	\$ 6,850	\$ 14,200
Mean Working Years	3.2 years	10.5 years	22.4 years
Dominant Attrition Rate	High (38%)	Low (8%)	Very Low (4%)



K-Means Clustering (K=3) - Employee Segments



5. Algorithmic Pseudocode

5.1 Proposed J48 Classification & Rule Extraction Algorithm

```
=====
ALGORITHM 1: J48_Attrition_Decision_Tree(Dataset D, Attribute_Set A)
=====
```

Input : Dataset D (2500 Instances), Candidate Attributes A

Output : Decision Tree T with Attrition Rules

```
1: Create a node N;
2: If all instances in D belong to class C (Attrition = Yes or No):
3:   Return N as a leaf node labeled with class C;
4: If Attribute_Set A is empty:
5:   Return N as a leaf node labeled with majority class in D;
6: For each attribute a in A:
7:   Compute Information Gain Gain(D, a) using Entropy equations;
8: Select best_attribute = argmax(Gain(D, a));
9: Set node N.test_attribute = best_attribute;
10: For each unique value v of best_attribute:
11:   Let D_v be the subset of D where best_attribute = v;
12:   If D_v is empty:
13:     Attach a leaf node labeled with majority class in D to N;
14:   Else:
15:     Attach subtree generated by J48_Attrition_Decision_Tree(D_v, A \
{best_attribute}) to N;
16: Perform Error-Based Pruning on Tree T using Confidence Factor C = 0.25;
17: Return Pruned Tree T;
=====
```

```
ALGORITHM 2: Naive_Bayes_Classification(Instance X)
```

```
1: For each class C_i in {Attrition=Yes, Attrition=No}:
2:   Calculate Prior Probability  $P(C_i) = \text{Count}(C_i) / \text{Total\_Instances}$ 
3:   Calculate Likelihood  $P(X | C_i) = \text{Product of } P(x_j | C_i) \text{ for all}$ 
attributes  $x_j$ 
4:   Compute Posterior  $P(C_i | X) = P(C_i) * P(X | C_i)$ 
5: Return class C_k with  $\text{argmax}(P(C_i | X))$ 
```

```
ALGORITHM 3: K_Means_Workforce_Clustering(Dataset D, K=3)
```

```
1: Randomly initialize 3 cluster centroids {c1, c2, c3} from D
2: Repeat:
3:   Assign each employee instance x to nearest centroid c_j using Euclidean
Distance
4:   Recalculate centroids c_j as mean of all instances assigned to cluster j
5: Until centroids no longer change position (Convergence)
```

6. Conclusion and Future Scope

6.1 Conclusion

This paper presented a comprehensive data warehousing and mining framework for forecasting employee turnover. By executing an integrated pipeline across WEKA and Python on 2,500 enterprise records, we demonstrated how OLAP data staging combined with predictive algorithms yields strategic HR insights:

1. **Classification:** The J48 Decision Tree achieved superior classification accuracy (85.2%) compared to Naive Bayes (82.9%).
2. **Key Drivers:** Information Gain and Association Rule Mining proved that **OverTime work**, **Low Job Level (Level 1)**, and **Low Monthly Income** are the highest drivers of voluntary attrition.
3. **Segmentation:** K-Means clustering isolated a high-risk workforce segment consisting of junior employees aged under 28 with low tenure and low monthly income.

6.2 Actionable HR Policy Recommendations

- **OverTime Cap:** HR managers should implement policies limiting consecutive overtime hours for Job Level 1 employees.
- **Targeted Compensation Adjustment:** Adjust base salaries for entry-level roles ($\text{MonthlyIncome} < \$3,500$) to align with competitive market medians.
- **Early Retention Interventions:** Focus retention incentives on Cluster 0 (Junior Staff during their first two years of service).

6.3 Future Scope

Future enhancements to this research include:

1. **Real-time OLAP Integration:** Deploying automated streaming ETL pipelines that refresh data cubes dynamically from live enterprise databases.
2. **Advanced Deep Learning:** Implementing Recurrent Neural Networks (RNNs) and XGBoost models to capture longitudinal temporal changes in employee performance over time.
3. **Sentiment Analysis:** Combining structured numerical logs with unstructured employee feedback text using Natural Language Processing (NLP).

References

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